Naive Bayes Classifier



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What Will We Learn?

- Basic Probability
- Conditional Probability
- Bayes' Theorem
- Prior, Likelihood, Evidence and Posterior
- Conditional Independence
- Naive Bayes Classification Process
- Classification Example

What is Probability?

- Probability measures how likely an event is to occur.
- Probability ranges from 0 to 1.
  0 → Impossible
  1 → Certain
  0.5 → 50% chance
- Example: 7 red balls and 3 blue balls → P(Red) = 7/10 = 0.7

Probability Notation

- P(A) → Probability of event A
- P(B) → Probability of event B
- P(A and B) → Probability that A and B both occur
- Example: A = Student passes;
- P(A) = probability that the student passes.

What is Conditional Probability?

- Probability of an event when we already know that another event has occurred.

- Notation: P(A | B)
- Read as: “Probability of A given B.”

- The information after | is the condition we already know.

Conditional Probability Example

Class: 60 students
40 male, 20 female
10 female students wear glasses

Question: What is P(Glasses | Female)?
Only the 20 female students are considered.
P(Glasses | Female) = 10/20 = 0.5 = 50%

Conditional Probability Formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

Probability of A and B

Probability of B

Conditional Probability & Deriving Bayes' Theorem

Deriving Bayes' Theorem — just algebra

1. $P(A \cap B) = P(A|B) \times P(B)$ and $P(A \cap B) = P(B|A) \times P(A)$
2. Both equal $P(A \cap B)$, so: $P(A|B) \times P(B) = P(B|A) \times P(A)$
3. Divide both sides by $P(B)$ → Bayes' Theorem.

Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

Why this matters for classification

- Relabel A as "class" and B as "observed data / features."
- We rarely know P(class | data) directly — but P(data | class) and P(class) are easy to estimate by counting training examples.
- Bayes' Theorem lets us flip the direction we can measure into the direction we actually want to predict.

Prior, Likelihood & Posterior Probabilities

$$P(\text{class} \mid \text{data}) = P(\text{data} \mid \text{class}) \times P(\text{class}) \;/\; P(\text{data})$$

P(class)	**Prior**	Belief about the class before seeing any data — e.g. 40% of all emails historically were spam. Estimated by counting class frequency in training data.
P(data | class)	**Likelihood**	How probable this exact data is, assuming the class is true — e.g. how often the word "free" appears in spam emails specifically.
P(data)	**Evidence**	Overall probability of seeing this data, regardless of class. Same value for every class being compared, so it can be dropped as a constant.
P(class | data)	**Posterior**	Updated belief after seeing the data — the answer we want. We compute one posterior score per class and rank them.

Since P(data) is constant across classes, in practice we simplify to: Posterior ∝ Likelihood × Prior

From Bayes' Theorem to a Classifier

$$P(class_k \mid x_1,\ldots,x_n) = [\, P(x_1,\ldots,x_n \mid class_k) \times P(class_k) \,] \;/\; P(x_1,\ldots,x_n)$$

The decision rule: Maximum A Posteriori (MAP)

$$\text{predicted class} = \operatorname{argmax}_k \; [\, P(x_1,\ldots,x_n \mid class_k) \times P(class_k) \,]$$

Two problems this still leaves us with

- $P(x_1,\ldots,x_n \mid \text{class})$ requires modeling how every feature interacts with every other feature — the number of parameters explodes as features grow.
- Estimating that full joint distribution needs enormous amounts of training data to avoid overfitting.
- Naive Bayes solves both problems with one simplifying assumption — covered next.

The Naive Conditional-Independence Assumption

Naive Bayes shortcuts the hard joint-probability problem by assuming every feature is independent of every other feature, given the class:

$$P(x_1, x_2, \ldots, x_n \mid \text{class}) \approx P(x_1|\text{class}) \times P(x_2|\text{class}) \times \ldots \times P(x_n|\text{class})$$

⚠ Usually false in reality

- "Free" and "money" are correlated in spam — not independent.
- In medical data, symptoms often co-occur for a reason.
- The model still treats every feature as if it were a separate, unrelated vote.

✓ Still works remarkably well

- Makes the math tractable — just multiply per-feature probabilities.
- Needs far less training data than modeling full feature interactions.
- Classification only needs the correct class to score highest — not perfect probabilities — and ranking often survives the wrong independence assumption.

What is Naive Bayes?

- Naive Bayes is a supervised machine learning classification algorithm.
- It is based on Bayes' Theorem.
- It uses the conditional independence assumption.
- It calculates the probability of each possible class.
- The class with the highest probability becomes the prediction.

Example 1

Day	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$\langle Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong \rangle$$

Example 1 Solution

Day	*Outlook*	*Temperature*	*Humidity*	*Wind*	*PlayTennis*
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$\langle Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong \rangle$

$$P(PlayTennis = yes) = 9/14 = .64$$

$$P(PlayTennis = no) = 5/14 = .36$$

Outlook	Y	N
sunny	2/9	3/5
overcast	4/9	0
rain	3/9	2/5
Tempreature		
hot	2/9	2/5
mild	4/9	2/5
cool	3/9	1/5

Humidity	Y	N
high	3/9	4/5
normal	6/9	1/5
Windy		
Strong	3/9	3/5
Weak	6/9	2/5

Example 1 Solution

$$\langle Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong \rangle$$

$$v_{NB} = \underset{v_j \in \{yes,no\}}{\text{argmax}} P(v_j) \prod_i P(a_i|v_j)$$

$$= \underset{v_j \in \{yes,no\}}{\text{argmax}} P(v_j) \quad P(Outlook = sunny|v_j) P(Temperature = cool|v_j)$$

$$P(Humidity = high|v_j) P(Wind = strong|v_j)$$

$$v_{NB}(yes) = P(yes)\ P(sunny|yes)\ P(cool|yes)\ P(high|yes)\ P(strong|yes) = .0053$$

$$v_{NB}(no) = P(no)\ P(sunny|no)\ P(cool|no)\ P(high|no)\ P(strong|no) = .0206$$

$$v_{NB}(yes) = \frac{v_{NB}(yes)}{v_{NB}(yes)+v_{NB}(no)} = 0.205 \qquad v_{NB}(no) = \frac{v_{NB}(no)}{v_{NB}(yes)+v_{NB}(no)} = 0.795$$

Example 2

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

$$\boldsymbol{New\ Instance} = (\boldsymbol{Red}, \boldsymbol{SUV}, \boldsymbol{Domestic}) \rightarrow (\boldsymbol{Yes\ or\ No})$$

Example 2 Solution

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

$$p(Yes) = \frac{5}{10} = 0.5$$

$$p(No) = \frac{5}{10} = 0.5$$

Color	Yes	No
Red	3/5	2/5
Yellow	2/5	3/5

Type	Yes	No
Sports	4/5	2/5
SUV	1/5	3/5

Origin	Yes	No
Domestic	2/5	3/5
Imported	3/5	2/5

$$New\ Instance = (Red, SUV, Domestic)$$ **No**

$$P(Yes|New\ Instance) = p(Yes) * P(Color = Red|Yes) * P(Type = SUV|Yes) * P(Origin = Domestic|Yes)$$

$$P(Yes|New\ Instance) = \frac{5}{10} * \frac{3}{5} * \frac{1}{5} * \frac{2}{5} = \frac{3}{125} = 0.024$$

$$P(No|New\ Instance) = p(No) * P(Color = Red|No) * P(Type = SUV|No) * P(Origin = Domestic|No)$$

$$P(No|New\ Instance) = \frac{5}{10} * \frac{2}{5} * \frac{3}{5} * \frac{3}{5} = \frac{9}{125} = 0.072$$

$$P(No|New\ Instance) > P(Yes|New\ Instance)$$

THANK YOU